

Raspberry Pi 5 - 1GB Version: Must - See Before Use

matters need attention:

1. **Due to system memory limitations**, the following cases may cause system crashes when running: AI advanced course cases, MediaPipe-related cases, offline large model deployment cases, Jupyter environment cases, ROS+OpenCV, visualization tool cases such as RVIZ, and simulation cases.
2. Some cases of program crashes and system freezes in this version are **not product quality issues**, but are caused by insufficient hardware memory resources.
3. In the event of a system crash, you can check the system memory usage status by executing the `free -h` and `htop` commands to assist in locating the problem.
4. The online large model case cannot be equipped with a voice interaction module, so only the text interaction function can be experienced.
5. **Cases marked with "★" and "🚫" in the figure below**, After testing, it cannot run normally on this platform. This is for reference only. **The actual availability is subject to the operation status of the equipment.**

Raspberry Pi5 Course outline

01)Introduction to Raspberry Pi 5	1)Introduction to Raspberry Pi 5		08)ROS1 advanced tutorial	ROS1 basic course	18)ROS-rqt tool usage★	
	2)Raspberry PI system installation and backup				19)Topic message recording and playback★	
	3)Powering the Raspberry Pi 5				20)urdf model introduction★	
	4)Startup of Raspberry Pi 5				21)gazebo introduction★	
	5)Installation and introduction of shell and active radiator				22)ROS distributed communication	
	6)GPIO and 40-pin pins			ROS1 vision application	1)AR vision	
	7)Tips for using Raspberry Pi 5				2)AR QR code	
	8)Raspberry PI PCIe connector				3)ROS+Opencv basics	
	9)MIPI CSI DSI connector				4)Face recognition★	
	10)UART connector				5)harris corner detection★	
	11)Real-time clock				6)Target tracking algorithm★	
	12)Power over Ethernet (PoE) connector				7)Contour moment★	
02)Introduction to common	1)Official Raspberry PI system				8)Polygon outline★	
	2)Ubuntu				9)Discrete fourier transform algorithm★	
03)Basic settings of Raspberry Pi	1)Update and upgrade operating system				10)Edge detection algorithm★	
	2)Introduction to raspi-config tool				11)Face detection algorithm★	
	3)Config.txt file description				12)Optical flow detection algorithm★	
	4)Network configuration				13)Contour detection★	
	5)Remote access				14)General contour detection★	
	6)Transfer files remotely				15)Feature point tracking★	
	7)Set display resolution and rotation				16)HLS color filtering★	
	8)Set screen to sleep				17)Hough circle detection★	
	9)Get real-time temperature of Raspberry Pi				18)Hough linear detection★	
	10)Play audio and video				19)HSV color filtering★	
	11)Using USB camera				20)LK optical flow algorithm	
	12)Using MIPI camera				21)detection algorithm★	
04)Basic programming course	A-C language programming	1)C language environment construction			22)Phase dependent displacement detection★	
		2)Print Hello World			23)Image pyramid sampling algorithm★	
	B-Python programming	1)Python usage			24)RGB color filtering★	
		2)Use APT to install Python packages			25)Clear background detection algorithm★	
		3)Python virtual environment			26)Simplified optical flow algorithm★	
		4)GPIO Zero			27)Simple filter★	
		5)Pin output high and low levels			28)Threshold image processing★	
		6)Pin reads high and low levels			29)Watershed segmentation algorithm★	
		7)Pin output PWM			ROS2 basic course	1) Introduction to ROS2
	C-Jupyter lab programming	1)Jupyter Lab environment construction★				2)ROS2 install Foxy
		2)Basic use of Jupyter Lab★				3)ROS2 development environment
		3) Jupyter Widgets configuration tutorial★				4)ROS2 workspace
		4)Jupyter Widgets getting started tutorial★				5)ROS2 function package
Open Source CV environment	1)Open Source CV environment setup★	6)ROS2 node				
	Getting started with Open Source CV	1)Introduction to Open Source CV★				7)ROS2 topic communication
2)Read and display Open Source CV images★		8)ROS2 service communication				
3)Open Source CV image writing★		9)ROS2 action communication				
4)Open Source CV image quality★		10)ROS2 custom interface message				
5)Open Source CV pixel operation★		11)ROS2 parameter service case				
Open Source	1)Image resizing★	12>About ROS2 meta-function package				
	2)Cut the image★	13)ROS2 distributed communication				
	3)Image translation★	14)ROS2 DDS				
					09)ROS2 advanced tutorial	

05)Advanced AI course	CV Geometric Transformations	4)Image mirroring★ 5)Affine transformation★ 6)Rotate the image★ 7)Perspective transformation★		15)ROS2 time related API 16)ROS2 common command tools 17)ROS2 rviz2 use★ 18)ROS2 rqt toolbox★ 19)ROS2 launch file configuration★ 20)ROS2 recording and playback tool★ 21)ROS 2URDF model★ 22)Gazebo simulation platform★ 23)TF2 coordinate transformation★
	Text Picture Drawing Open Source CV image processing and drawing text line segments	1)Grey Processing★ 2)image binaryzation★ 3)edge detection★ 4)Line segment drawing★ 5)Rectangle and circle drawing★ 6)Text image drawing★		1)Opencv basic course★ 2)ROS+opencv application★ 3)QR code creation and recognition★ 4)AR vision★ 5)Interesting gameplay of media
	Advanced AI gameplay	1)QR code★ 2)Human pose estimation★ 3)target detection★ 4)yolov5 5)Camera display 6)face detection★ 7)Color recognition★		
	YOLOv11 Advanced Usage	1) YOLOv11 Environment Setup★ 2) CLI Usage★ 3) Object Detection★ 4) Instance Segmentation★ 5) Pose Estimation★ 6) Image Classification★ 7) Oriented Object Detection★ 8) Model Conversion★ 9) Dataset Annotation★ 10) Model Training and Conversion★ 11) Model Prediction★	Offline large model(Text Interaction)	1)AI large model environment deployment★ 2)Chinese input method switching★ 3)Large model dialogue platform installation★ 4)Meta AI Llama3.2 model★ 5)Alibaba Cloud Qwen3 model★ 6)Microsoft Phi-4-mini model🔗 7)DeepSeek DeepSeek-R1 model★ 8)Alibaba Cloud Qwen2.5v visual multimodal large model🔗 9)Google Gemma3 visual multimodal large model★ 10)Lavvam🔗 11)MiniCPM-V visual multimodal large model🔗 12)Multimodal visual understand application🔗 13)Multimodal text image application★ 14)Multimodal video analysis application🔗 15)Multi module visual position application🔗 16)mutimdal tabcanapln🔗 17)Multimodal autonomous agent application🔗
06)Mediapipe course		1)Face special effects 2)Face detection 3)3D object recognition★ 4)Paintbrush★ 5)Finger control★ 6)Gesture recognition★		1)Offlinespeech totext (ASR)🔗 2)Offline text to speech (TTS)★ 3)AI large model voice interaction 4)Multimodal visual understand speech interaction🔗
	07)Docker Tutorial	1)Overview and installation 2)Common commands 3)Understand and publish images 4)Hardware interaction data processing 5)Enter docker container 6)Update docker images 7)docker container to install RO	Offline large model(Voice Interaction)	5)Mutimdal text and graphic application★ 6)Multimodal video analysis application🔗 7)Multi module visual position application🔗 8)Multimodal table scanning application🔗 9)Multimodal autonomous proxy application🔗 10)AI large model offline voice assistant🔗
08)ROS1 advanced tutorial	ROS1 basic course	1)ROS introduction 2)ROS installation 3)ROS common command tools 4)ROS workspace 5)ROS function package 6)ROS node 7)ROS topic publisher 8)ROS topic subscribers 9)ROS service client 10)ROS service server 11)ROS action client	Online large model(Text Interaction)	1)OpenRouter large model API aggregation platform 2)Multimodal visual understand application 3)Multimodal Text-to-Image Generation Application 4)Multimodal Video Analysis Application 5)Multimodal visual localization application 6)Multimodal table scanning application 7)Multimodal autonomous proxy application
				1)Online speech to text (ASR) 2)Online text to speech (TTS) 3)AI large model voice interaction 4)Multimodal visual speech interaction

	12)ROS action server	Online large model (Voice Interaction)	5)Multimodal Text-to-Image Generation Application
	13)ROS custom message reception		6)Multi module visual position application
	14)ROS-launch file		7)Multimodal Video Analysis Application
	15)ROS-TF transformation★		8)Multimodal table scanning application
	16)ROS parameter service★		9)Multimodal autonomous proxy application
	17)ROS-rviz use★		10)AI large model offline voice assistant
Due to performance limitations, the Raspberry Pi 5-1GB version cannot use cases with "★" and "🌀", and the Raspberry Pi 5-2GB/4GB version cannot use cases with "🌀".			